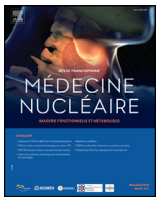
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Original article

Harmonization of exposure constraints for relatives following targeted radionuclide therapy: A French perspective

Harmonisation des contraintes d'exposition pour les proches à la suite d'une radiothérapie interne vectorisée : une perspective française

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ABSTRACT

Targeted radionuclide therapy is being used more and more to treat both benign and malignant pathologies. It is therefore highly desirable to harmonize care practices, particularly with regard to radiation protection rules for relatives and carers. Based on literature data and consensus among the Radiation Protection Group of the SFMN, the proposed methodology uses standardized scenarios and situations for patients and their relatives. When applied to real treatment conditions, it shows low restriction durations for patients treated for thyroid cancer (upon exiting iratherapy chambers) and for LUTATHERA treatments. These durations are particularly significant for patients with benign thyroid diseases and prostate cancer, as well as for pregnant women, young couples and children. Based on a small number of hypotheses and initial data, this work proposes harmonising recommendations for restriction duration through the use of freely available, internet-based software for the entire nuclear medicine community. This software has the potential to be adapted to new radiopharmaceuticals and regulatory situations.

Mots clés:
Radiothérapie interne vectorisée
Règles de radioprotection
Situation particulière de l'exposition de l'entourage
Recommandations

RÉSUMÉ

Du fait de l'utilisation actuelle croissante de la radiothérapie interne vectorisée en pathologies bénignes et malignes, une harmonisation des pratiques de soins est souhaitable, en particulier en ce qui concerne le respect des règles de radioprotection de l'entourage et des aidants des patients. A partir des données de la littérature et d'un consensus établi au sein du groupe Radioprotection de la SFMN, la méthodologie présentée se base sur des scénarii standardisés tout en pouvant s'adapter à la situation particulière de l'entourage d'un patient donné. L'application aux conditions de traitement réelles montre que si ces durées sont très faibles pour les patients traités pour cancer de la thyroïde (à la sortie de chambre d'irathérapie) ou pour les traitements de type LUTATHERA, elles sont plus importantes pour le traitement des maladies bénignes de la thyroïde ou des cancers de la prostate, en particulier pour les conjointes enceintes, les conjoints jeunes ou les enfants en bas âge. En se basant sur un minimum d'hypothèses et de données initiales, ce travail propose une harmonisation des recommandations de durée d'éviction, à travers d'un logiciel librement disponible pour l'ensemble de la communauté de la Médecine nucléaire, tout en pouvant s'étendre à de nouveaux radiopharmaceutiques ou de nouvelles situations réglementaires.

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1. Introduction

Since the early 1940s, targeted radionuclide therapy (TRT) using iodine-131 has been used to treat a large number of patients with malignant and benign thyroid conditions (with approximately 15,000 treatments carried out in France each year in 2024) [1]. The treatment of liver cancer by selective internal radiation therapy is steadily growing in France, with around 1000 treatments carried out in 2024. In rheumatology, the use of radioactive particles labelled with yttrium-90, rhenium-186 or erbium-169 is also common practice in many countries.

However, the development of new TRT radiopharmaceuticals around fifteen years ago using both beta and alpha emitters has renewed interest in TRT for treating common diseases, particularly prostate cancer. The number of treatment with ^{177}Lu-PSMA-617 in France increased from virtually zero in 2021 to approximately 4600 doses in 2024 (for around 2500 patients) [1]. Clinical trials [2,3] are currently underway to treat patients at an earlier stage of their disease, which should increase the number of patients eligible for these treatments even further. In addition, other therapeutic radiopharmaceuticals are in development for widespread neoplastic diseases [4]. This rapid increase in the use of TRT for common conditions raises numerous logistical and organizational questions, requiring adjustments to healthcare provision in terms of suitable premises, trained staff and the corresponding financial resources.

Given the large number of patients involved, and in the absence of any proven risks to those around them when they leave healthcare facilities, most of these treatments are carried out during short hospital stays, known as outpatient care lasting less than 24 hours [5]. However, measurable activity in patients upon discharge exposes relatives to ionizing radiation from low-level external irradiation. This is often perceived by the public (and even those around the patients) as posing a significant risk, even though the doses delivered are always well below the threshold for the first known effects in humans (100 mSv).

The ICRP Publication 94 [6] has proposed specific dose constraints for the general public, taking into account the radiosensitivity of different life stages (e.g., embryogenesis, early childhood, adulthood, senior citizens over 60, etc.), with wide safety margins. These recommendations are transposed into national legislation, taking into account the healthcare needs of the population and the availability of healthcare facilities. In France, the generally adopted dose constraints for caregivers are 1 mSv for embryos (over nine months, i.e., for pregnant women) and children (up to 18 years of age, 1 mSv per year), 3 mSv for adults up to 60 years of age (excluding pregnancy), and 15 mSv for those over 60.

In practice, it is necessary to inform family and friends of patients undergoing TRT of the precautions required to minimize their exposure, while ensuring they can still provide medical care, comfort and companionship. In addition to the risk of contamination (both external and internal) and the management of potentially contaminated protective equipment, this involves defining how long restrictions on close contact should last. Historically, recommendations for these periods have varied greatly, ranging from no restrictions to very long fixed periods. The latter are not justified from a radiation protection perspective for relatives and completely impractical for patients, particularly elderly patients who require constant care and regular psychological and relational support. For instance, the latest survey conducted in 2024 by the French Society of Nuclear Medicine (SFMN) among 34 nuclear medicine departments revealed that 37% instructed patients to store urine-contaminated disposable waste at home for 0–8 days post-treatment. In this survey, 24% of departments recommended storing the waste for 9–15 days or 16–30 days. A quarter recommended a period of more than 30 days. This results in considerable heterogeneity in waste management and recommended disposal periods between healthcare facilities, sometimes even within the same city. This leads to patient confusion regarding these varying recommendations.

That is why the Radiation Protection group of the SFMN, a multidisciplinary team comprising medical physicists (French Society of Medical Physicists [SFPM]), radiopharmacists (French Society of Radiopharmacy [SoFRa]) and technologists (French Society of Nuclear Medicine Technologists [AFTMN]), has proposed a methodology that can be applied to most common situations in the patient's life. This methodology harmonises recommendations on exclusion periods at a national level.

The objectives of this work are twice. First, it aims to standardize the recommendations on exclusion periods given to patients' families and friends by proposing standardized recommendations with the option of creating a specific scenario for a patient's particular environment. Second, it proposes software that is easily accessible to all departments and recommends exclusion periods based on the radiopharmaceutical used and exposure at 1 metre.

2. Materials and methods

2.1. General formalism

The formalism is based on a model proposed in 2004 [7], whose purpose is to calculate exclusion times based on effective half-lives adapted to the radiopharmaceutical, exposure scenarios and dose constraints. Briefly, the calculation of the effective dose received by relatives after leaving the hospital sector depends on a contact scenario that is repeated daily at a variable distance. Below, we present the general case for a calculation based on a two-distance scenario.

$\dot{D}_T^{\min}$ is the minimum effective dose rate that leads to the maximum effective dose $D_T^{\max}$. Distances d_1 and d_2 are those between the patient and the contact subject considered in the model and on which the scenarios will be based. $\dot{D}_T^{\min}$ follows (times are in days):

$$\dot{D}_T^{\min} = D_T^{max} \frac{1}{f(d_1) + kf(d_2)}$$

With k, the factor to convert dose rate from d_2 to d_1. The function f follows:

$$f(d) = \frac{T_{eff}}{\ln 2}\left(1-\exp\left(\frac{-\ln 2}{T_{eff}}\Delta t\right)\right)\frac{1}{1-\exp\left(\frac{-\ln 2}{T_{eff}}\right)}$$

With Δt the contact time between the patient and the contact subject per day and T_{eff} the effective half-life related to the injected radiopharmaceuticals. The factor k follows the relation proposed by Siegel [8]:

$$k = \frac{\tan^{-1}\frac{l}{2d}}{l\frac{d}{2}}$$

With l the patient's size and d the distance between the patient and relatives.

The minimum time $t_{\lim}$ required to comply with the instructions leading to $D_T^{\max}$ is therefore given by:

$$t_{\lim} = \frac{T_{eff}}{ln2}\ln\frac{\dot{D}_T^{\min}}{\dot{D}_C}$$

With D the dose rate measured on the day of discharge.

This approach is considered for all radiopharmaceutical treatments except for the treatment of benign thyroid diseases with iodine-131, where measuring the dose rate at one metre immediately after administration is not very meaningful. Instead, an estimate of the dose rate is made based on the iodine uptake rate (usually estimated beforehand by scintigraphy, ideally using iodine-123), the activity administered and the dose rate constant for iodine-131 in the thyroid [9].

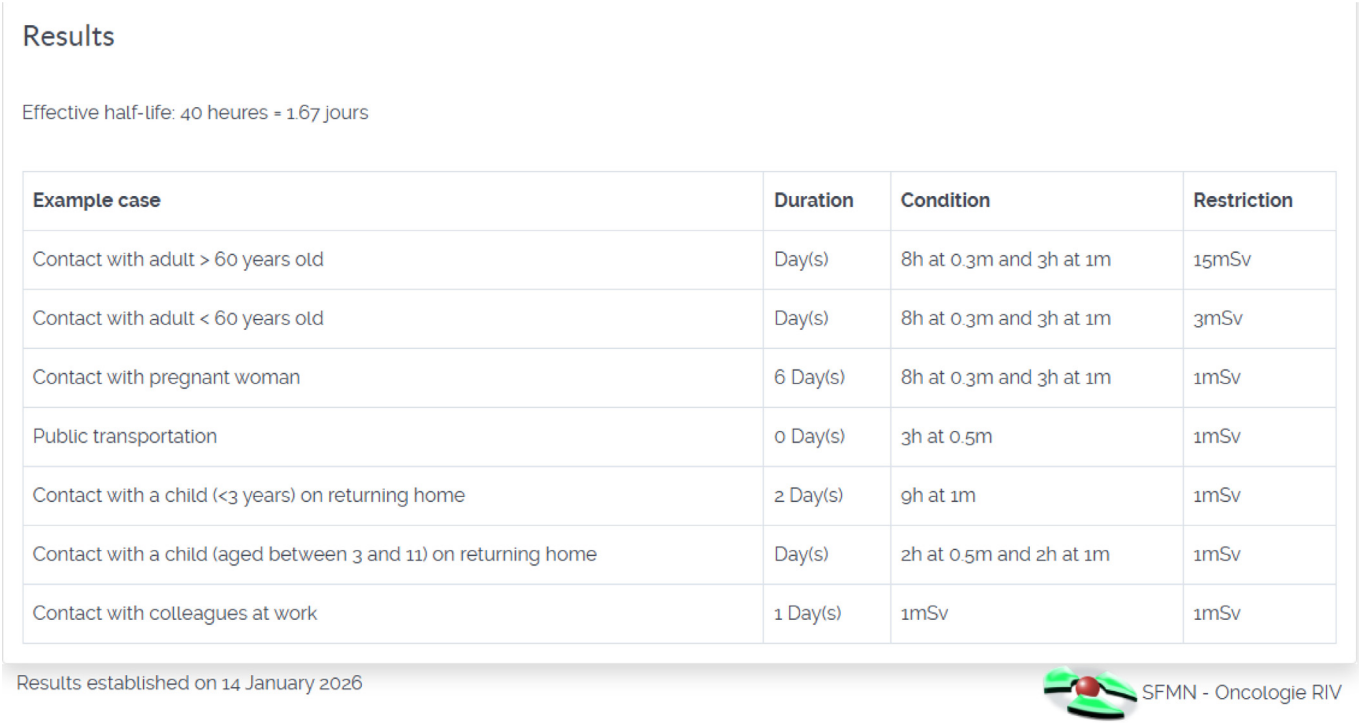


Fig. 1. Illustration of the output from the online tool for a patient who has undergone six cycles of 177Lu-PSMA and has a dose rate of 20 μSv/h upon leaving hospital. The patient is 170 cm tall.Illustration des résultats de l'outil en ligne pour un patient ayant reçu six cycles de 177Lu-PSMA et un débit de dose de 20 μSv/h à sa sortie de l'hôpital. Le patient mesure 170 cm.

2.2. Choice of scenarios, dose limits and effective half-lives

Several practical situations involving fixed, unchangeable exposure (Fig. 1) were initially considered based on Barrington's proposals [10]. These were then modified following a discussion involving professionals from various disciplines within the SFMN Radiation Protection working group. The aim of these modifications was to approximate the most realistic exposure conditions while deliberately selecting maximal exposure times. However, a free scenario is allowed in order to approximate exposure conditions not found in the fixed scenarios. The scenarios can be found in Fig. 1, alongside with the results of a treatment involving six cycles of ^{177}Lu-PSMA. Note that the final output can be inserted into the patient's record.

The scenarios are set with a maximum effective dose limit $D_T^{\max}$ in accordance with the recommendations of a European Community report [11] and following collegial and multidisciplinary discussion. These dose constraints are set for one year and are adjusted according to the number of ^{177}Lu-DOTATATE and ^{177}Lu-PSMA-617 treatment courses, or any other treatment requiring multiple injections.

The mean effective half-lives used are taken from data in the literature for each radiopharmaceutical and are listed in Table 1.

2.3. Assessment based on practical cases

The effectiveness of this approach was evaluated based on four types of therapy: ^{177}Lu-DOTATATE, ^{177}Lu-PSMA-617, differentiated thyroid cancer and benign thyroid conditions (Table 2). Measurements were taken on the day of hospital discharge (six hours after the injection of ^{177}Lu-DOTATATE and ^{177}Lu-PSMA-617, and three days after administration of the iodine-131 capsule). Four treatments were considered for ^{177}Lu-DOTATATE and six for ^{177}Lu-PSMA-617, regardless of the actual number received by the patient.

Table 1
Effective half-lives [7,12–18].
Demi-vies effectives [7,12–18].

Radiopharmaceuticals	Effective half-life (h)	Reference
Iode-131-Cancer	16.00	Carlier, et al. [7]
Iode-131-Benign disease	122.40	Carlier, et al. [12]
PSMA-177Lu	40.00	Kratochwil, et al. [13]
Radium-223	274.32	Physical period
NET-177Lu	100.00	Fitschen, et al. [14,15]
Microspheres-90Y	64.05	Physical period
Microspheres-166Ho	26.81	Physical period
Lipiodol-131I	192.96	Giammarile, et al. [16]
MIBG 131I	30.60	Wafelman, et al. [17]
Synovectomy-90Y	64.08	Clunie, et al. [18]
Synovectomy-186Re	88.80	Clunie, et al. [18]
Synovectomy-169Er	225.60	Clunie, et al. [18]

3. Results

The main characteristics of the treatments considered are given in Table 2.

The exclusion times are shown in Tables 3–6 for ^{177}Lu-DOTATATE, ^{177}Lu-PSMA-617, ^{131}I cancer and ^{131}I benign, respectively.

Table 7 shows the recommended exclusion periods for benign thyroid diseases based on the uptake percentage for a fixed administered activity of 740 MBq. In practice, high uptake values (above 50%), which almost exclusively occur in cases of highly active Graves' disease, are rare and not treated with high administered activities. Consequently, the vast majority of cases fall within the upper half of the table.

4. Discussion

The overall objective of TRT is to provide effective treatment for patients while minimising exposure to ionizing radiation for relatives

Table 2
Main characteristics of the different treatments.
Principales caractéristiques des différents traitements.

Treatment	Number of treatments	Mean injected activity (MBq)	Mean dose rate at 1 m (μSv/h)
^{177}Lu-DOTATATE	60	7153 [6754–7541]	13.3 [7–20]
^{177}Lu-PSMA-617	197	7287 [6037–7637]	18.2 [10–31]
^{131}I cancer	23	3355 [1100–3806]	8.0 [1–22]
^{131}I bénin	121	564 [307–777]	

Table 3
Contact restriction for patients treated with ^{177}Lu-DOTATATE with discharge from the hospital ward 6 hours after injection (60 treatments).
Restriction des contacts pour les patients traités au ^{177}Lu-DOTATATE avec sortie du service hospitalier 6 heures après l'injection (60 traitements).

Scenario	Adult > 60 years	Non-pregnant adult < 60 years	Pregnant woman	Public transportation	Child < 3 years	Child 3–11 years	Co-worker
No restriction (%)	100	22	0	99	82	95	100
1 day of restriction (%)	0	67	1	1	18	5	0
2 days of restriction (%)	0	12	17	0	0	0	0
3 days of restriction (%)	0	0	70	0	0	0	0
≥ 4 days of restriction (%)	0	0	12	0	0	0	0

Table 4
Contact restriction for patients treated with ^{177}Lu-PSMA-617 with discharge from the hospital ward 6 hours after injection (197 treatments).
Restriction des contacts pour les patients traités par ^{177}Lu-PSMA-617 avec sortie du service hospitalier 6 heures après l'injection (197 traitements).

Scenario	Adult > 60 years	Non-pregnant adult < 60 years	Pregnant woman	Public transportation	Child < 3 years	Child 3–11 years	Co-worker
No restriction (%)	98	0	0	100	0	2	9
1 day of restriction (%)	2	0	0	0	2	14	87
2 days of restriction (%)	0	2	0	0	71	82	4
3 days of restriction (%)	0	58	0	0	26	2	0
4 days of restriction (%)	0	39	0	0	1	0	0
5 days of restriction (%)	0	1	8	0	0	0	0
6 days of restriction (%)	0	0	88	0	0	0	0
≥ 7 days of restriction (%)	0	0	4	0	0	0	0

Table 5
Contact restriction for patients treated with ^{131}I for differentiated thyroid cancer with discharge after 3 days (23 treatments).
Restriction des contacts pour les patients traités par l'iode 131 pour un cancer différencié de la thyroïde avec sortie après 3 jours (23 traitements).

Scenario	Adult > 60 years	Non-pregnant adult < 60 years	Pregnant woman	Public transportation	Child < 3 years	Child 3–11 years	Co-worker
No restriction (%)	100	100	100	100	100	100	100

Table 6
Contact restriction for patients treated with ^{131}I for benign conditions (121 treatments, measured uptake values from 2.9 to 49%, mean 13.4%).
Restriction des contacts pour les patients traités avec de l'iode 131 pour des affections bénignes (121 traitements, valeurs d'absorption mesurées de 2,9 à 49 %, moyenne 13,4 %).

Scenario	Adult > 60 years	Non-pregnant adult < 60 years	Pregnant woman	Public transportation	Child < 3 years	Child 3–11 years	Co-worker
No restriction (%)	100	94	41	99	99	99	100
1 day of restriction (%)	0	1	12	1	0	1	0
2 days of restriction (%)	0	2	10	0	0	0	0
3 days of restriction (%)	0	1	8	0	1	0	0
4 days of restriction (%)	0	1	8	0	0	0	0
5 days of restriction (%)	0	0	6	0	0	0	0
6 days of restriction (%)	0	1	2	0	0	0	0
7 days of restriction (%)	0	0	5	0	0	0	0
≥ 7 days of restriction (%)	0	0	7	0	0	0	0

and for society as a whole during the decay period. This period appears to be short (a few hours to a few days), especially compared to conventional therapeutic tools such as chemotherapy and antibiotic therapy whose drug molecules and metabolites can persist in the environment for a long time due to their limited biodegradability [19]. This article proposes the use of a calculator to standardize practices for recommending exclusion periods when patients who have undergone TRT return home or to an institution.

Recommendations for radiation protection for patients undergoing TRT are provided in the form of specific written and/or verbal information, and sometimes during a dedicated consultation. These recommendations are based on local, national or international guidelines. However, practices remain heterogeneous, probably due to the heterogeneous nature of the recommendations themselves. The IAEA recommends a maximum exposure dose constraint for patients' families and caregivers [20]. For family members and friends, these limits are 1 mSv per treatment for children aged 3 to 10, 3 mSv for adults under 60 and 15 mSv for adults over 60. For external caregivers, the limit is 0.3 mSv per treatment. In other cases, the limit of 1 mSv per year applies. These recommendations are adapted from ICRP 94 [6]. The US Nuclear Regulatory Commission recommends a maximum exposure of 5 mSv per treatment for family members who are exposed [9]. Specific recommendations for patients' families and friends are not always included in radiopharmaceuticals' summaries of product characteristics (SmPCs), and those that are often lack a documented scientific basis. For example, the SmPC for iodine-131

Table 7
Number of days of contact restriction for patients treated with ^{131}I for benign conditions based on the percentage of uptake (for an administered activity of 740 MBq).
Nombre de jours de restriction de contact pour les patients traités avec du ^{131}I pour des affections bénignes en fonction du pourcentage d'absorption (pour une activité administrée de 740 MBq).

Iodine uptake (%)	Adult > 60 years	Non-pregnant adult < 60 years	Pregnant woman	Public transportation	Child < 3 years	Child 3–11 years	Co-worker
5%	0	0	0	0	0	0	0
10%	0	0	3	0	0	0	0
20%	0	0	8	0	0	0	0
30%	0	3	11	0	0	0	0
40%	0	5	13	0	1	0	0
50%	0	6	14	2	3	1	0
60%	0	8	16	3	4	3	0
70%	0	9	17	4	5	4	1
80%	0	10	18	5	7	5	2
90%	0	11	19	6	7	6	3
100%	0	11	19	7	8	6	4

marketed in France does not provide any specific recommendations. The SmPC for ^{177}Lu oxodotreotide (Lutathera®) specifies that, after each administration, close contact with other people should be limited to less than1 metre for seven days. For children and/or pregnant women, this should be limited to less than 15 minutes per day for seven days. Patients must also sleep in a separate room for 7 days; this period is extended to 15 days for children and/or pregnant women. Finally, the SmPC contains no specific recommendations for ^{177}Lu-PSMA-6 17 (Pluvicto®).

In comparison with other recently developed software with a similar purpose [21–23], the proposed approach is intended for everyday use in various clinical departments. This requires a simple interface that is easy to understand and edit for integration into patient records, while ensuring the protection of everyone in the vicinity and covering all radiopharmaceuticals available in routine clinical practice.

The various selected parameters were the subject of extensive discussion within the SFMN group, based on international references and recommendations. In most cases, radiopharmaceuticals are located at multiple sites within the body (for example, multiple bone metastases in the case of ^{177}Lu-PSMA), resulting in multiple scattered sources. This explains the choice of a linear source model. Moreover, the single-exponential model was chosen for these calculations to reflect real-life exposure conditions (outside healthcare facilities), which, in most cases, only occur a few hours or days after administration of the radiopharmaceutical (for example, after hospitalization and upon returning home). Under these conditions, the initial phase of rapid distribution and elimination has mostly passed, leaving only the slow elimination component. Moreover, the effective half-lives used are based on mean values found in the literature and strike a good balance between protecting those around the patient and the fear of isolation, which can be a barrier to treatment acceptance by elderly patients (who are most likely to be affected by TRT).

Note that in cases where excretion of the radiopharmaceutical is slow (e.g., radium-223, holmium-166 microspheres, etc.), and therefore its effective half-life is close to its physical half-life, the latter has been used as the maximum value for precautions (with no significant change to the proposed exclusion periods).

The proposed "standardized" scenarios aim to replicate most exposure situations for relatives while remaining adapted to normal conditions upon returning home. The advantage of these fixed scenarios is that they allow for consistent exposure constraints for relatives that are not dependent on the Nuclear Medicine department. While multiple scenarios have been proposed in the literature, some are clearly not very representative. For instance, the case of a patient who has recently undergone treatment and, despite recommendations to limit contact, carries a new-born baby 30 centimetres from an irradiating organ (such as the thyroid) for eight hours a day, does not seem to be representative of the population treated with TRT. These standardized scenarios have been the subject of lengthy debate within the SFMN Radiation Protection Group in recent years and represent a relative consensus on the subject. Where the situation upon returning home differs significantly from that envisaged in the "standardized" scenarios, the "patient" scenario must be used and adapted to reflect the specific circumstances. For instance, the situation of a single caregiver (whether personal or paramedical) providing care over prolonged periods (usually less than an hour per day in close proximity to the patient) after returning home or to an institution can be modelled by adjusting the exposure parameters (duration and distance) to reflect this care situation as closely as possible.

Please note that the present methodology concerns only the external irradiation exposure and that contamination risks (external and internal ones) require preventive procedures following recommendations like SFMN ones [24].

Finally, the ability to adjust the effective half-life means this approach can be used for cohorts of volunteer patients in clinical Phase I/II research, for whom a standardized period for the radiopharmaceutical and pathology in question has not yet been published. This applies to trials using alpha emitters or new β^- radioelements, for example.

The online tool is available at: https://www.acoramen.org/eviction.

5. Conclusion

This approach, which is designed to be user-friendly and covers the most common conditions of exposure to others after TRT, has been made available to the nuclear medicine community. It can adapt to the various specific situations that may be encountered in clinical practice. It harmonizes these recommendations for exclusion periods across all territories subject to French legislation, while remaining adaptable to other regulatory situations.

Disclosure of interest

The authors declare that they have no competing interest.

References

[1] https://www.cnp-mn.fr/wp-content/uploads/2025/03/20250323_ENQUETE_3.0_SITE_SFMN-.pdf.
[2] Eapen RS, Buteau JP, Jackson P, Mitchell C, Oon SF, Alghazo O, et al. Administering [177Lu]Lu-PSMA-617 prior to radical prostatectomy in men with high-risk localised prostate cancer (LuTectomy): a single-centre, single-arm, phase 1/2 study. Eur Urol. 2024;85:217–26.
[3] Kishan AU, Valle LF, Wilhalme H, Felix C, Nabong R, Juarez-Casillas JE, et al. 177Lu-prostate-specific membrane antigen neoadjuvant to stereotactic ablative radiotherapy for oligorecurrent prostate cancer (LUNAR): an open-label, randomized, controlled, phase II study. J Clin Oncol. 2025;43:3812–21.

[4] Tran HH, Yamaguchi A, Manning HC. Radiotheranostic landscape: a review of clinical and preclinical development. Eur J Nucl Med Mol Imaging. 2025;52:2685–709.
[5] ASN-lettre-circulaire-12-juin-2020-évolutions-recommendations-lutetium-177-Mai-2017.
[6] ICRP Publication 94. Release of patients after therapy with unsealed radionuclides. Ann ICRP. 2004;34:1–79.
[7] Carlier T, Lisbona A, Kraeber-Bodere F, Ansquer C, Couturier O. Recommandations pratiques concernant la sortie des patients après traitement du cancer différencié de la thyroïde à l'131I. Radioprotection. 2004;39:481–92.
[8] Siegel JA, Kroll S, Regan D, Kaminski MS, Wahl RL. A practical methodology for patient release after tositumomab and (131)I-tositumomab therapy. J Nucl Med. 2002;43:354–63.
[9] NRC Regulatory Guide 8.39 (Rev.2, 2021) https://www.nrc.gov/docs/ML2123/ML21230A318.pdf.
[10] Barrington SF, Kettle AG, O'Doherty MJ, Wells CP, Somer EJR, Coakley AJ. Radiation dose rates from patients receiving iodine-131 therapy for carcinoma of the thyroid. Eur J Nucl Med. 1996;23:123–30.
[11] European commission. Radiation protection following iodine-131 therapy (exposures due to outpatients or discharged inpatients). Radioprotection. 1999;97, https://op.europa.eu/en/publication-detail/-/publication/cf28c3bc-dff4-4602-95dd-1d0bfb7c8fcc.
[12] Carlier T, Salaun PY, Cavarec MB, Valette F, Turzo A, Bardiès M, et al. Optimized radioiodine therapy for Graves' disease: two MIRD-based models for the computation of patient-specific therapeutic 131I activity. Nucl Med Commun. 2006;27:559–66.
[13] Kratochwil C, Fendler WP, Eiber M, Baum R, Bozkurt MF, Czernin J, et al. EANM procedure guidelines for radionuclide therapy with 177Lu-labelled PSMA-ligands (177Lu-PSMA-RLT). Eur J Nucl Med Mol Imaging. 2019;46:2536–44.
[14] Fitschen J, Knoop BO, Behrendt R, Knapp WH, Geworski L. External radiation exposure and effective half-life in Lu-177-Dota-Tate therapy. Z Med Phys. 2011;21:266–73.
[15] Levart D, Kalogianni E, Corcoran B, Mulholland N, Vivian G. Radiation precautions for inpatient and outpatient 177Lu-DOTATATE peptide receptor radionuclide therapy of neuroendocrine tumours. EJNMMI Phys. 2019;6:7.
[16] Giammarile F, Bodei L, Chiesa C, Flux G, Forrer F, Kraeber-Bodere F, et al. Therapy, Oncology and Dosimetry Committees. EANM procedure guideline for the treatment of liver cancer and liver metastases with intra-arterial radioactive compounds. Eur J Nucl Med Mol Imaging. 2011;38:1393–406.
[17] Wafelman AR, Nortier YLM, Rosing H, Maessen HJM, Taal BG, Hoefnagel CA, et al. Renal excretion of meta-iodobenzylguanidine after therapeutic doses in cancer patients and its relation to dose and creatinine clearance. Nucl Med Commun. 1995;16:767–72.
[18] Clunie G, Fischer M, EANM. EANM procedure guidelines for radiosynovectomy. Eur J Nucl Med Mol Imaging. 2003;30:BP12–6.
[19] IAEA. Release of patients after radionuclide therapy, 67. Vienna: IAEA Safety Reports Series; 2009 [p. 1–77].
[20] Castellano-Hinojosa A, Gallardo-Altamirano MJ, González-López J, Gonzáles-Matinez A. Anticancer drugs in wastewater and natural environments: a review on their occurrence, environmental persistence, treatment, and ecological risks. J Hazard Mater. 2023;447:130818.
[21] Broggio D, Célier D, Michel C, Isambert A. Contact restriction time after common nuclear medicine therapies: spreadsheet implementation based on conservative retention function and individual measurements. J Radiol Prot. 2023;43:021504.
[22] Forster JC, Badger D, Hickson KJ. Close contact restriction periods for patients who have received iodine-131 therapy for differentiated thyroid cancer. J Radiol Prot. 2023;43:021501.
[23] Otis G. A solution for iterative determination of patient release instructions for fractionated radionuclide therapy. Health Phys. 2020;119:766–71.
[24] Fiche d'information à l'usage des patients bénéficiant d'une administration thérapeutique d'Iode radioactive. https://www.cnp-mn.fr/wp-content/uploads/2024/05/consignes131-I-V1.0.pdf.